

Product Family Declaration

Family Name: WiFi Interface Converter Series

Representative Model : **RS485-WiFi-004 (RS485 Interface)**

Core Hardware: Based on ESP32-C3-MINI-1-N4 module with 5V USB/Terminal power and integrated LDO.

Variants:

- Model CAN-WiFi-003: Features CAN 2.0B interface with integrated transceiver (TCAN3414). Supports transparent data transmission between CAN bus and Wi-Fi
- Model UART-WiFi-001: UART interface (GPIO pins for TX/RX)
- Model SPI-WiFi-002: SPI interface (GPIO pins for MOSI/MISO/SCK/CS/REQ)
- Model **RS485-WiFi-004**: Dual RS485 interface (utilizing **SP3485EN-L/TR** transceiver, A/B parallel bus, 10k bias, 120Ω termination, TVS protection), supports **Slave & Transparent Master mode**.

Hardware Consistency : All models in this series share the same RF circuitry, Wi-Fi antenna, power supply regulation (LM1117-3.3V), and PCB core layout. The only differences are the peripheral interface pins and corresponding interface chips (e.g., CAN transceiver vs. RS485 driver).

Compliance: All variants utilize the pre-certified RF module **FCC ID: 2AC7Z-ESP32C3MINI1**. No changes have been made to the RF characteristics or antenna integration.

Manufacturer: WNetTech

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